

CSE357:COMBINATORIAL STUDIES

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamental computer science concepts, including data structures, algorithms, databases, operating systems, and computer networks, essential for technical interviews.

CO2 :: assess the problem-solving skills specific to coding challenges and algorithmic problems frequently encountered in technical interviews.

CO3 :: articulate a comprehensive command of Object-Oriented Programming principles, enhancing readiness to excel in technical interviews.

CO4 :: gain comprehensive strategies and techniques to effectively prepare for technical interviews, including mock interviews, resume building, and effective communication during interviews.

CO5 :: impart the requisite knowledge and skills to confidently address technical interview inquiries posed by service-oriented IT companies, demonstrating proficiency across diverse technical domains.

CO6 :: prepare and practice sessions including mock interviews, whiteboard coding exercises, and simulations of behavioural interview scenarios, fostering confidence and preparedness for different stages of technical interviews.

Unit I

Operating System Basics : Foundations of Operating Systems, Types of Operating Systems, Memory Management, Processor Scheduling Algorithms, Process Synchronization and Inter-process Communication

Unit II

Computer Networking Basics : Foundations of Computer Networks,, Network Topologies and Transmission Medium,, Understanding OSI and TCP/IP Models,, Subnetting and Routing,, Key Protocols: HTTP, SMTP, POP/IMAP, FTP, DNS, DHCP,

Unit III

Database Management Systems (DBMS) : Introduction to Databases and RDBMS,, Data Definitions: Tables, Fields, Records,, SQL and Data Manipulation,, Database Keys and Data Integrity,, Database Normalization and Transactions,, Transaction Management in DBMS

Unit IV

Fundamentals of Programming Languages : C/C++/Java Interview Questions Overview,, Pointers and Storage, Classes,, Introduction to Object-Oriented Programming (OOP),, Techniques of Parameter Passing and Binding,, Memory Handling in OOP Languages

Unit V

Data Structures : Introduction to Data Structures, Linked Lists (Types and Operations), Stacks and Queues (Operations and Applications), Trees(Binary Trees and Balanced Trees), Graphs (Representation and Algorithms), Hashing (Applications and Functions), Heaps(Applications and Operations)

Unit VI

Algorithms : Understanding Algorithms and Analysis, Running Time Analysis and Rate of Growth, Asymptotic Notation: Big-O Notation, Recursion and Backtracking, Dynamic Programming Strategies, Greedy Algorithms

List of Practicals / Experiments:

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- 1. Introduction to Linux Commands
- 2. Shell Programming
- 3. System Calls (File / Directory/ Process)
- 4. Multi Thread Process using Pthread Library
- 5. SQL Commands (DDL/DML/ TCL)

- 6. PL/SQL Triggers, Procedures, Functions
- 7. Subnetting
- 8. Arrays and Linked List
- 9. Stacks and Queues
- 10. Trees and Binary Search Trees

Text Books: 1. IT INTERVIEW QUESTIONS by NARASIMHA KARUMANCHI, CAREERMONK PUBLICATIONS

References:

1. MCQS IN COMPUTER SCIENCE by TIMOTHY WILLIAMS, MC GRAW HILL
2. SQL IN 10 MINUTES, SAMS TEACH YOURSELF by BEN FORTA, SAMS PUBLISHING
3. CRACKING THE IT INTERVIEW by M BALASUBRAMANIAM, K. R BAALAJI, KIRAN. G RANGANATH, NANDAWAT RAVINDRA K, M SELVAGURU, T COMERICA SUBASH, RAGHAVAN S VENKAT, S ANBAZHAGAN VIKRAM, MC GRAW HILL
4. CRACKING THE CODING INTERVIEW by MCDOWELL GAYLE LAAKMANN, CAREERCUP